Sleep, Stress & Lifestyle Behavior Analysis

SQL & Python-Based Behavioral Data Analysis

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The Problem We Set Out to Solve

Daily Habits & Sleep

How do daily behavioral choices — screen time, physical activity, sleep routines — actually affect sleep quality and duration?

Behavioral Relationships

What measurable connections exist between phone usage before bed, activity levels, and reported stress scores?

High-Risk Patterns

Can we identify a segment of users whose combination of habits puts them at elevated risk for poor sleep and chronic stress?

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Dataset Overview

User-level behavioral and lifestyle data capturing real-world daily habits across a diverse population.

Sleep Duration

Hours of sleep per night

Stress Level

Self-reported 1–10 scale

Physical Activity

Frequency and intensity of exercise

Occupation

Professional category per user

Screen & Phone Usage

Daily screen time + pre-sleep phone use

Demographics

Gender breakdown across user base

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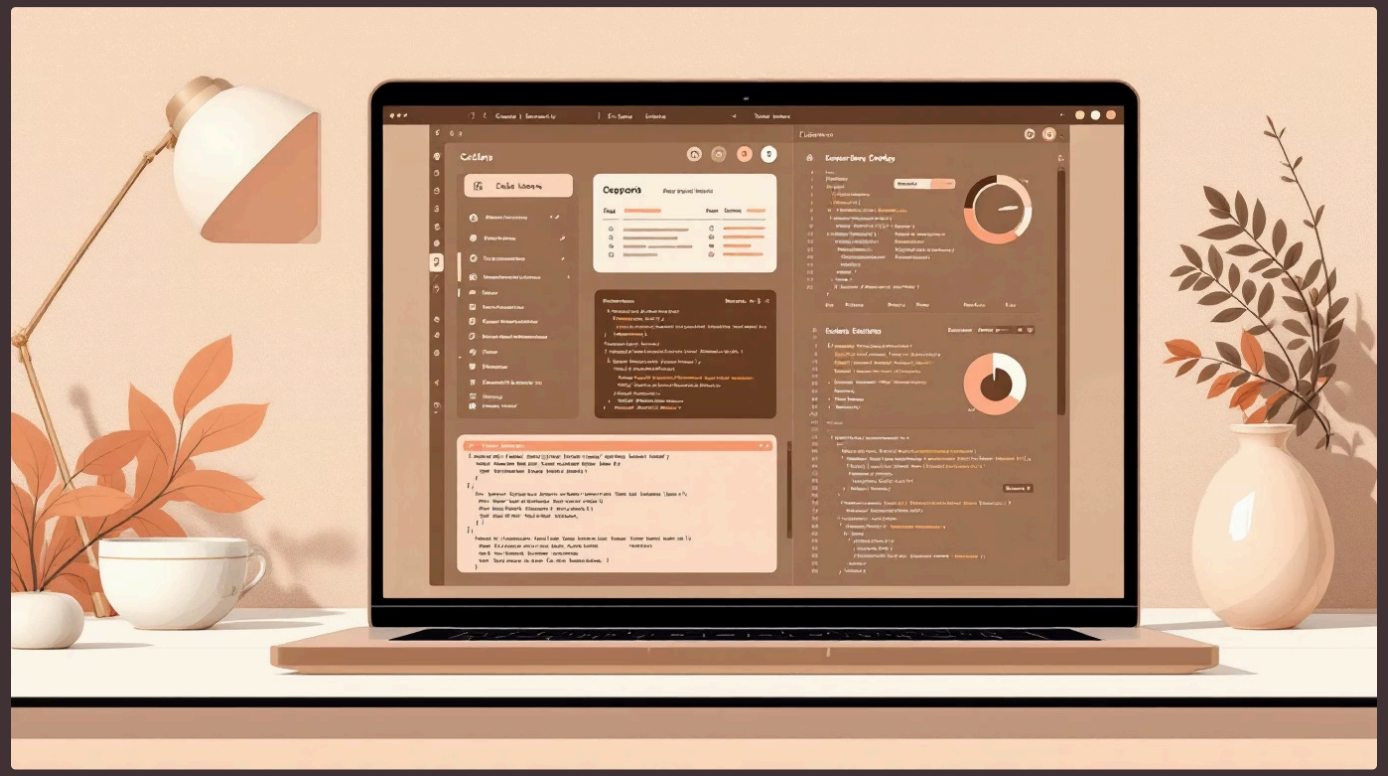
Data Preparation & Cleaning



Making the Data Analysis-Ready

Before any analysis could begin, the dataset required careful validation and preparation to ensure reliable results.

- Scanned for **missing or inconsistent values** and resolved data quality issues
- Validated and enforced **correct data types** for numeric and categorical fields
- Structured the dataset for seamless use in both **SQL queries and Python-based analysis**

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Feature Engineering & Categorization

Raw numeric values were transformed into structured behavioral categories — enabling meaningful group comparisons across the dataset.

Screen Time

Low / Moderate / High daily usage buckets

Pre-Sleep Phone Use

Categorized by usage intensity before bedtime

Physical Activity

Low / Moderate / High activity level tiers

Categorization unlocks structured cross-group comparisons that raw numeric data alone cannot reveal.

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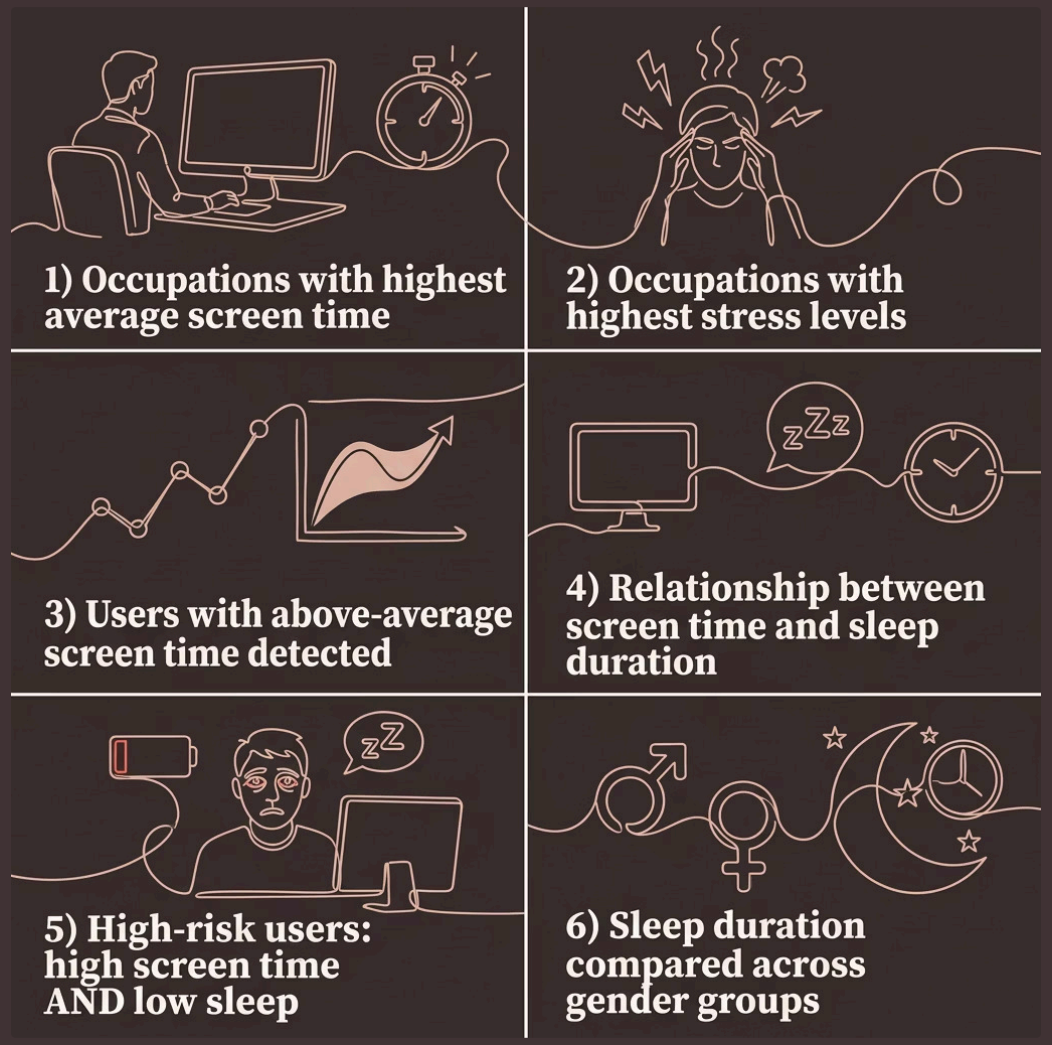
SQL Analysis — Key Queries & Findings



What SQL Revealed

Structured queries surfaced behavioral patterns invisible in raw data — from occupation-level stress profiling to identifying a high-risk user segment combining excessive screen time with dangerously low sleep.

High-risk users — those with above-average screen time AND below-average sleep — emerged as a distinct, actionable segment.

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Python Analysis — Exploratory Deep Dive

Exploratory Data Analysis

Profiled distributions, detected outliers, and summarized behavioral trends across the full dataset

Correlation Analysis

Measured statistical relationships between screen time, physical activity, sleep duration, and stress scores

Visualizations

Box plots and distribution charts made patterns and outliers immediately apparent and interpretable

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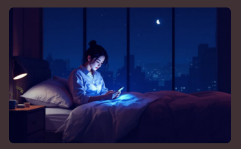
KEY FINDINGS

What the Data Told Us



Screen Time → Less Sleep

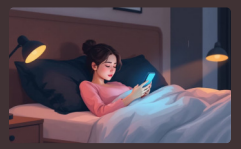
Higher daily screen time consistently correlated with shorter sleep duration across all user groups.

Activity → Lower Stress

Users with higher physical activity levels reported significantly lower stress scores.

Occupation Matters

Certain professions showed consistently elevated stress, pointing to role-specific behavioral risk factors.



Pre-Sleep Phone Use

Phone usage immediately before sleep negatively impacted perceived sleep quality across the dataset.

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Real-World Impact & Applications

From Data to Action

These insights extend well beyond the dataset — they map directly onto tools and decisions that affect daily wellbeing.

🏥 Health & Wellness Apps

Personalized nudges based on screen time and activity patterns

⚠️ Early Risk Detection

Flag high-risk behavioral profiles before stress or sleep issues escalate

📊 Lifestyle Awareness

Data-driven evidence to support healthier habit formation decisions

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Tools, Takeaways & Thank You

Tools & Technologies

SQL — aggregation, filtering, grouping, subqueries

Python — Pandas, EDA, correlation analysis

Jupyter Notebook — exploration and visualization

Conclusions

This project demonstrated that **lifestyle habits are measurably linked to sleep quality and stress** — and that structured data analysis can make those links visible and actionable.

- Behavioral data yields meaningful, human-relevant insights
- SQL and Python together enable robust exploratory analysis
- High-risk patterns can be identified and acted upon early

Thank You

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